



Flash ESP32 Firmware Flashing Guide



Flash Download Tool 3.9.6

Dalian Good Display Co., Ltd.

1. Document Overview

This guide provides detailed instructions on flashing firmware to the ESP32 microcontroller using Espressif Flash Download Tool Version 3.9.6. The tool supports ESP32 chip series and primarily performs flashing via the UART interface. It is applicable to both development (Develop mode) and mass production (Factory mode) scenarios. Certain features exclusive to Version 3.9.6 (e.g., the chipInfoDump tab) may differ slightly from those of later versions (e.g., $\geq 3.9.8$), while the core flashing functionality remains fully compatible.

Note: Back up all important data before flashing to prevent accidental flash memory erasure. Ensure the firmware files are in .bin format.

2. Preparation

Hardware Requirements

- 1 ESP32E6-E01 or ESP32133-C02 module (for firmware flashing)
- 1 PC running Windows 7 (64-bit) or Windows 10
- 1 USB-to-UART adapter (for connecting to the ESP32 UART interface)

Software Requirements

Flash Download Tool Version 3.9.6 (launch by double-clicking the .exe file)

Prepared firmware files

(in .bin format, e.g., bootloader, partition table and application firmware)

3. Tool Interface Introduction

The main interface will pop up after launching the tool (double-click the .exe file). The key elements are as follows:

ChipType: Select the chip model

(ESP32S3 is selected in this guide).

WorkMode: Select the working mode

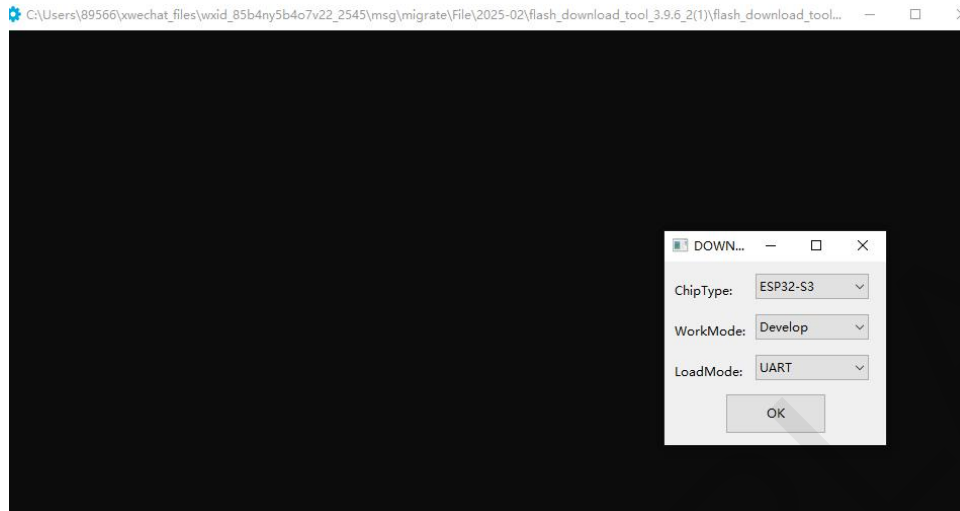
Develop mode: Uses absolute file paths; suitable for single-chip development and flashing.

Factory mode: Uses relative file paths (it is recommended to place firmware files in a bin folder in the same directory as the tool's .exe file). The interface is locked by default.

Click LockSettings to unlock editing.

Configuration settings will be saved locally when the tool is closed.

LoadMode: Only supports the UART communication mode.



After selecting an option, click OK to enter the corresponding tab page:

- Develop mode: Enters the SPI Download tab page.
- Factory mode: Enters the FactoryMultiDownload tab page.
- SPIDownload Tab Page (Develop Mode)
- It is used for standard firmware flashing in the development environment and is mainly divided into three sections:
- Download Path Config: Configure firmware paths and download addresses (in hexadecimal format, e.g., 0x1000). Tick the checkbox preceding the path to include the corresponding file. Multiple files can be added (e.g., bootloader at 0x1000, partition table at 0x8000, and application firmware at 0x10000).
- SPI Flash Config:
- SPI SPEED: Set the SPI boot speed (e.g., 40 MHz; common values: 20 MHz, 40 MHz or 80 MHz, depending on the flash chip model).
- SPI MODE: Set the SPI boot mode (e.g., DIO, QIO; QIO is the default mode for ESP32).
- DETECTED INFO: Automatically displays the detected flash and crystal oscillator information after the flashing process starts.
- DoNotChgBin: When enabled, the tool flashes the raw content of the .bin file directly; when disabled, the firmware will be updated according to the configured SPI SPEED and MODE before flashing.
- CombineBin Button: Merges all selected firmware files into a single file (saved as ./combine/target.bin). If DoNotChgBin is enabled, the raw files will be merged; otherwise, the merged file will be generated with the configured SPI parameters applied. Unused regions between files will be filled with 0xff. Each click on this button will overwrite the previously generated merged file.

SPIDownload Tab Page (Develop Mode)

Designed for standard firmware flashing in development environments, it is mainly divided into three sections:

Download Path Config: Configure firmware file paths and download addresses (in hexadecimal format, e.g., 0x1000). Tick the checkbox before each path to include the corresponding file. Multiple files can be added (e.g., bootloader at 0x1000, partition table at 0x8000, and application firmware at 0x10000).

SPI Flash Config:

SPI SPEED: Set the SPI boot speed (e.g., 40 MHz; common values: 20 MHz, 40 MHz or 80 MHz, depending on the flash chip type).

SPI MODE: Set the SPI boot mode (e.g., DIO, QIO; QIO is the default mode for ESP32).

DETECTED INFO: Automatically displays the detected flash and crystal oscillator information after the flashing process starts.

DoNotChgBin: When enabled, the tool flashes the raw content of the .bin file directly; when disabled, the firmware will be updated based on the configured SPI SPEED and MODE before flashing.

CombineBin Button: Merges all selected firmware files into a single file (saved as ./combine/target.bin). If DoNotChgBin is enabled, the raw files will be merged; otherwise, the merged file will be generated with the configured SPI parameters applied. Unused regions between files will be filled with 0xff. Each click will overwrite the previously generated merged file.

Default Button: Restore SPI configuration to default values (standard settings for ESP32).

Download Panel:

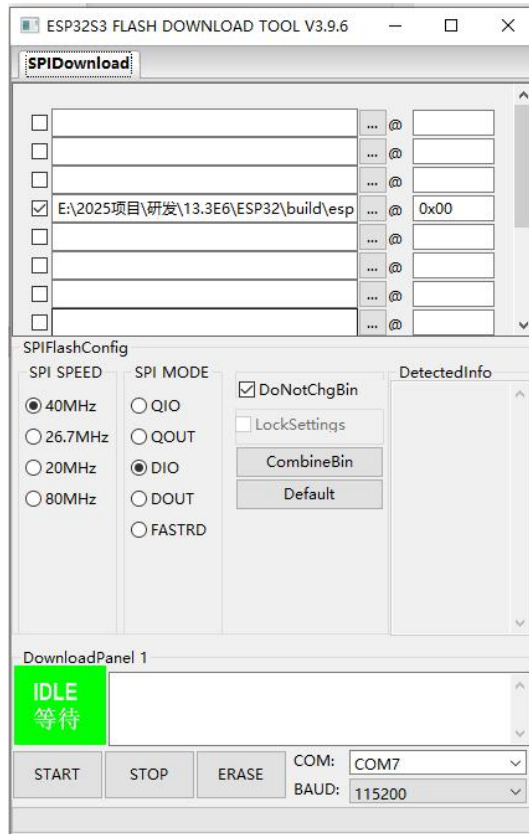
START: Initiate the firmware flashing process.

STOP: Terminate the flashing process.

ERASE: Erase the entire flash memory (use with caution to avoid data loss).

COM: Select the serial port (e.g., COM3).

BAUD: Set the baud rate (e.g., 115200; higher baud rates such as 921600 can accelerate the flashing speed, but a stable connection is required).



Note: Our company will provide a pre-packaged BIN file. Therefore, only add this pre-packaged BIN file, fill in the offset address as 0x00, set the correct serial port and baud rate, then click START to start the flashing process.

4. Flashing Process

Standard Download (ESP32 Example)

1. Complete the configurations as specified above.
2. In the SPIDownload or FactoryMultiDownload tab page, click START.
3. The tool establishes a connection via UART, detects the flash memory information and MAC address, and then flashes the firmware to the specified address.
4. Monitor the progress on the interface. Once the process is completed, the message "FINISH" will be displayed.